How the three ensembles come to agree

An exact study of a 2D quantum lattice gas

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The question

A gas can be described three ways:

- **microcanonical** — fix energy and particle number
- **canonical** — fix temperature, exchange energy with a bath
- **grand-canonical** — also exchange particles

They agree *in the thermodynamic limit*. Any finite system — a trap, a simulation — is not there yet.

What we measure

How the agreement sets in as the lattice grows, and how interactions change it.

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Model and method

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The model: hard-core Bose–Hubbard

$$\hat{H} = -t \sum_{\langle i,j \rangle} \left(\hat{b}_i^\dagger \hat{b}_j + \text{h.c.} \right) + V \sum_{\langle i,j \rangle} \hat{n}_i \hat{n}_j$$

- bosons hop between neighbors (t)
- repel or attract neighbors (V)
- at most one boson per site

- cold atoms in an optical lattice
- energy in units of t; $k_B = 1$

Not the quantum lattice-gas *automaton* — same name, different thing.

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Exact diagonalization

- $\hat{H}$ conserves N: block diagonal, one block per sector
- diagonalize each block *fully* — the partition sums need every level
- sizes $L = 2, 3, 4$ ($M = 4, 9, 16$ sites); half-filled block is the cost ceiling

One spectrum $\{E_{N,\alpha}\}$, built once and cached, feeds all three ensembles.

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Ensembles and what to measure

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Three ensembles, one spectrum

Microcanonical

$$\Omega(E, N; \Delta E)$$

$$S = \ln \Omega$$

Canonical

$$Z_N = \sum_{\alpha} e^{-\beta E_{N,\alpha}}$$

$$C_V = \beta^2 \mathrm{Var}(E)$$

Grand

$$\Xi = \sum_{N} e^{\beta \mu N} Z_N$$

$$\partial_\mu \langle N \rangle = \beta \, \mathrm{Var}(N)$$

Each is a different weighting of the same levels.

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Equivalence is a fluctuation

Canonical weight is a Laplace transform of the count:

$$Z_N = \int dE\, e^{S(E)-\beta E}.$$

The saddle at $S'(E^*) = \beta$ ties the two temperatures together; its width gives

$$\frac{\sigma_E}{|\langle E\rangle|} \sim \frac{1}{\sqrt{M}}, \qquad \frac{\sigma_N}{\langle N\rangle} \sim \frac{1}{\sqrt{M}}.$$

These ratios are the small parameters of equivalence — and smooth spectral sums, so they survive on a tiny spectrum.

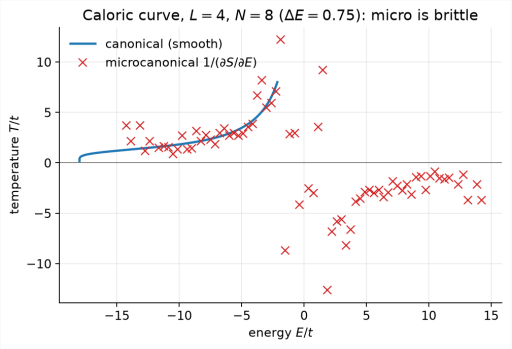
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Why not read T off the entropy?

$1/T = \partial S/\partial E$ is the honest definition — and useless here.

- $S = \ln \Omega$ is a jagged histogram
- slope diverges in gaps
- flips sign past the band center (negative T)

So we track the fluctuations instead.



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Does it compute the right thing?

Checked against independent references:

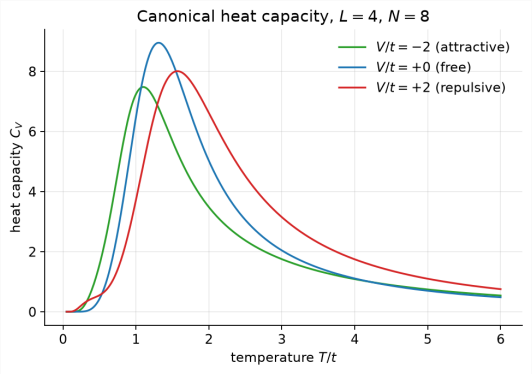
- one-particle band matches $-2t[\cos k_x + \cos k_y]$ to 10^{-15}
- ground states match a sparse Lanczos solver
- $C_V = \beta^2 \mathrm{Var}(E)$, $\partial_\mu \langle N \rangle = \beta \mathrm{Var}(N)$ vs numerical derivatives (10^{-5})
- $T \to 0, \infty$ limits of S and C_V
- exact limits of $\sigma_N / \langle N \rangle$: binomial as $\beta \to 0$; $\sqrt{(M+2)/3M}$ at $V = -2t$, $T \to 0$ (to 10^{-9})

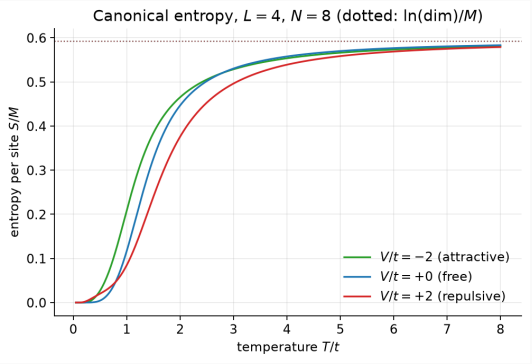
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Results

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Thermodynamics behaves

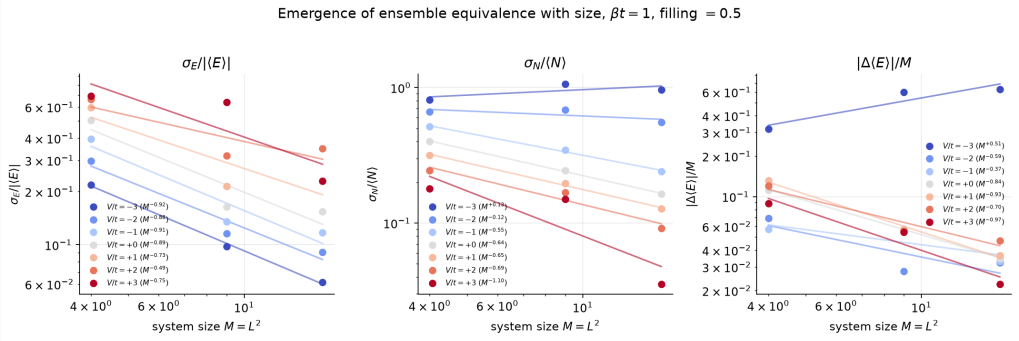




Schottky-like heat capacity; entropy climbs from 0 to $\ln(\dim)/M$. Interaction shifts the scale, not the shape.

Dhruv

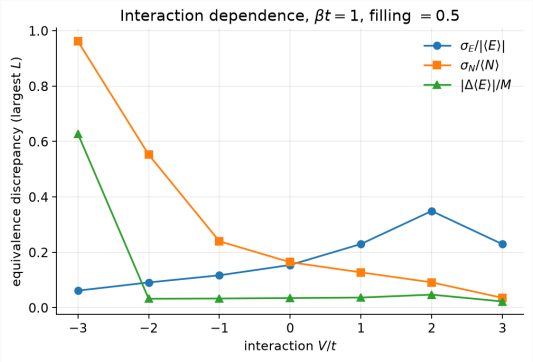
Equivalence emerges with size



Free and repulsive: all three discrepancies fall with M, near $M^{-1/2}$.

Dhruv

Attraction breaks it — in the number channel



V/t	$p(\sigma_N/\langle N\rangle)$	$\sigma_N/\langle N\rangle$
-3	$+0.13$	0.96
0	-0.64	0.16
$+3$	-1.10	0.04

At $V/t = -3$ the exponent is **positive**: the mismatch *grows* with size — the gas wants to phase separate.

All at one temperature, $\beta t = 1$. Is this physics, or a finite-size accident?

Dhruv

What theory predicts for the number channel

No hopping ($t = 0$): $\hat{n}_i = \frac{1+s_i}{2}$ gives an Ising model, $J = |V|/4$. Onsager:

$$T_c \approx 0.567\,|V| \quad (1.70\,t \text{ at } V = -3t)$$

Below T_c: dense *or* dilute, so $\sigma_N/\langle N\rangle \to m_0(T)$ — **size independent**.

Hot ($\beta \to 0$): binomial, exactly $M^{-1/2}$.

With hopping: $\hat{b}^\dagger = \hat{S}^+$ gives an XXZ ferromagnet,

$$J_z = |V|, \qquad J_{xy} = 2t.$$

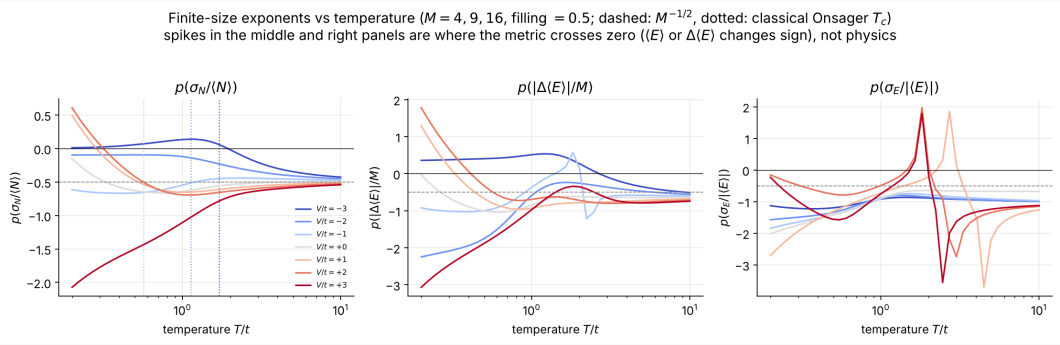
- $V < -2t$: easy axis ⇒ phase separation
- $V = -2t$: Heisenberg point, $T_c = 0$
- $-2t < V < 0$: easy plane, superfluid

Prediction

The anomaly needs $V < -2t$ *and* T below a T_c that is at most $0.567|V|$.

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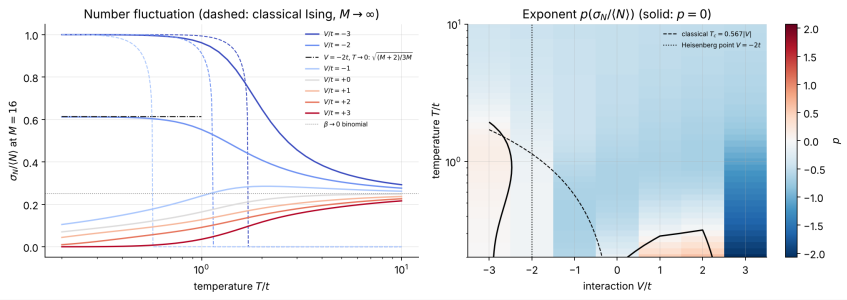
Adding a temperature axis



Fit $M = 4, 9, 16$ at 40 temperatures, $T/t = 0.2$–10. Hot: every coupling heads to $-1/2$ (binomial gives -0.49 on these sizes). Cold: attraction pulls the exponent up toward 0, and only $V/t = -3$ crosses it.

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The anomaly lives where theory puts it



- $V/t = -3$: exponent > 0 only below $T \approx 1.9\,t$; classical $T_c = 1.70\,t$
- $V/t = -2$: never positive; plateaus at the Heisenberg value $\sqrt{18/48} = 0.612$
- signature: exponent stuck near 0 rather than $-1/2$ (either sign)

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Takeaway

The ensembles converge for free and repulsive particles, and fail to for attractive ones — in the particle number, where phase separation lives: below its T_c, and only for $V < -2t$.

Measured through fluctuations, because the microcanonical thermometer fails on small bounded spectra.

Thank you

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Hard-core Bose–Hubbard model, exact diagonalization